

Spring Hackathon

Outcomes

- Understand what Docker is and run a containerised web app
- Explore a React frontend and how it communicates with a Python backend
- Train and visualise a model of your choice on a timeseries problem

Important information

- This is not a "competitive" hackathon
- Have fun with it
- Limit AI use

Tools

What is Flask?

- A lightweight Python web server.
- It listens for requests from the browser and sends back data or pages.

```
@app.route("/")
```

```
def index():
```

```
    return render_template("index.html")
```

What is React?

A JavaScript library for building user interfaces.

Instead of manually updating the page, you just update your data and React updates the page automatically.

```
const [results, setResults] = useState([]);
```

When results changes, the charts re-render – no extra code needed.

What is Docker?



- A tool that packages your app and everything it needs to run into a single container.
- Think of it like a box – open it on any machine and it just works.
- Two main files
 - A Dockerfile – instructions for building the container
 - A docker-compose.yml – shortcut to run it

Dockerfile



- A Dockerfile – instructions for building the container

```
# Use an official lightweight Python image
FROM python:3.11-slim

# Set the working directory inside the container
WORKDIR /app

# Copy and install dependencies first (Docker caches this layer if requirements.txt hasn't changed)
COPY requirements.txt .
RUN pip install -r requirements.txt

# Copy the rest of the app
COPY . .

# Tell Flask to be visible outside the container (0.0.0.0 instead of localhost)
ENV FLASK_RUN_HOST=0.0.0.0

# Run the app
CMD ["python", "app.py"]
```

docker-compose.yml



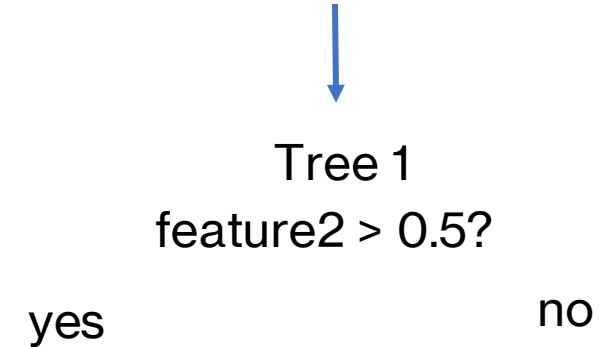
- A docker-compose.yml – shortcut to run it

```
1  services:
2    webapp:
3      build: .          # build from the Dockerfile in this folder
4      ports:
5        - "5000:5000"   # map localhost:5000 to the container's port 5000
6
```

What is XGBoost?

- A powerful machine learning algorithm that builds many small decision trees, each one learning from the mistakes of the last.
- The result is a strong model built from many weak ones (known as boosting).
- One of the most popular algorithms for structured data.

Sample: [feature0, feature1, feature2, feature3, feature4]



The data

Domain Co	Domain	Area Code	Area	Element Co	Element	Item Code	Item	Year Code	Year	Unit	Value
QC	Crops	2	Afghanistan	5419	Yield	56	Maize	1961	1961	hg/ha	14000

year	country	avg_temp
1849	Côte D'Ivoire	25.58

Area	Year	average_rain_fall_mm_per_year
Afghanistan	1985	327

Domain	Area	Element	Item	Year	Unit	Value
Pesticides	Albania	Use	Pesticides	1990	tonnes of active ingredients	121

Direction?

Does the amount of pesticide work as a good input for yield forecasting?

Can we use temperature to help predict a countries rainfall, can this predicted data then be used as an input for a yield prediction model?

Does a model trained on global data perform better at predicting country X's yield rather than a model trained only on country X?

Is the above effected by model choice?

Groups!

Group 1 Jack Davis Athinoulla Konstantinou James Nelson Videsh Jagroo Seyi Ibigbemi Poppy Ferguson	Group 2 Dominik Bittner Becca Morrell-Tomiczek Yiyuan Wang Jack MacKenzie John Callaghan Georgie Fletcher
Group 3 Aura Anderson-Ross Alisa Holm Arkan De Lomas Abena Nkansah Albor Meha	Group 4 Villanelle O'Reilly Lauren Gilman James Barnard Maddison Parkhouse Adam Hale

Example Code

The repo:

[SUSTAIN-CDT/TimeseriesHackathon: Timeseries Hackathon example project.](#)

How to run it

- Install docker on your computer
- [Windows | Docker Docs](#)
- Make sure Docker is running
- Navigate to the folder in your console and run:
- `docker compose up --build`
- In your browser go to `http://localhost:5000/`

If I was you...

- Study the example code
- Create new routes and functions to test you understand how the app functions
 - Online documentation for Flask and React are fantastic
- Get the provided data working with the provided XGBoost model
- XGBoost has no temporal awareness by default. It treats every row as an independent sample. Look at how it can be windowed, how does this affect performance
- Investigate other models Random Forest, LSTM, Timeseries transformer models, Moirai.

The outcome

- By the end of the week, we expect
 - The Web app using the provided data to create forecasts of any kind
 - Visuals of the results presented on the web app
- Any thing you try to explore which doesn't work, or you didn't finish, tell us!
 - It's interesting to hear the story and challenges faced